

Project Name:

AI-Powered Email Management Agent

Created with:

n8n (Self-hosted), Google Gemini, Gmail Integration

Purpose:

Automate email-related tasks using an AI Agent that can understand user requests, check emails, classify them, add labels, and send emails.

Logic Flow Breakdown:

• Chat Trigger:

- Starts the workflow when a user enters a request through the chat interface.
- The user can request actions such as sending an email or checking recent emails.

• AI Agent:

- Acts as the central controller of the workflow.
- Understands the user's request and decides which tool should be used.

• Google Gemini LLM:

- Provides intelligence to the AI Agent.
- Helps understand user requests and analyze email content.

• Gmail Tool – Get Emails:

- Retrieves recent emails from the connected Gmail account.
- Provides email information to the AI Agent for analysis.

• Email Classification:

- Emails are classified into three categories:
 - URGENT
 - NORMAL
 - SPAM

- **Gmail Tool – Add Label:**

- Adds appropriate labels to emails.
- URGENT emails receive an URGENT label.
- SPAM emails receive a SPAM label.
- NORMAL emails do not receive any additional label.

- **Gmail Tool – Send Email:**

- Sends emails based on the user's instructions.
- The AI Agent extracts the recipient, subject, and message and uses the Gmail Send tool.

Tested Use Cases:

Input:

Send an email to avimurkute.516@gmail.com with the subject "Test" and message "Testing Gmail connection."

Output:

The AI Agent successfully uses the Gmail Send tool and sends the email.

Input:

Check my most recent email and classify it as URGENT, NORMAL, or SPAM.

Output:

The AI Agent retrieves the email using Gmail, analyzes it using Gemini AI, classifies it, and performs the appropriate action.

Dependencies / Config Needed:

- **Gmail credentials connected through n8n OAuth.**
 - **Google Gemini API credentials.**
 - **Gmail account with required permissions.**
 - **n8n instance (Self-hosted or Cloud).**
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Ideas for Future Improvements:

- Add persistent memory for conversation context.
- Automatically summarize long emails.
- Add priority scoring for emails.
- Send notifications for urgent emails.
- Support multiple Gmail accounts.
- Add calendar integration for meeting-related emails.
- Create automatic replies for selected emails.